CSC490 Assignment A3: Pre-training nanochat

Prithvi Seran (1009973343) Vraj Shah (1009102695) Tyseer Toufiq (1009364064)
Joshua Crisologo (1009860438)

Part 1: Architecture Review

The Original nanochat (October 13, 2025)

The original nanochat [1] (commit `3a5e0bc`) is a decoder-only, pre-norm Transformer [2, 3]. Its size is set by the depth L alone: the width is $d = 64L$, each head has 128 dimensions, and the MLP hidden size is $4d$. The d20 model used for the speedrun (Figure 1) has about 561M parameters. Table 1 lists its main design choices.

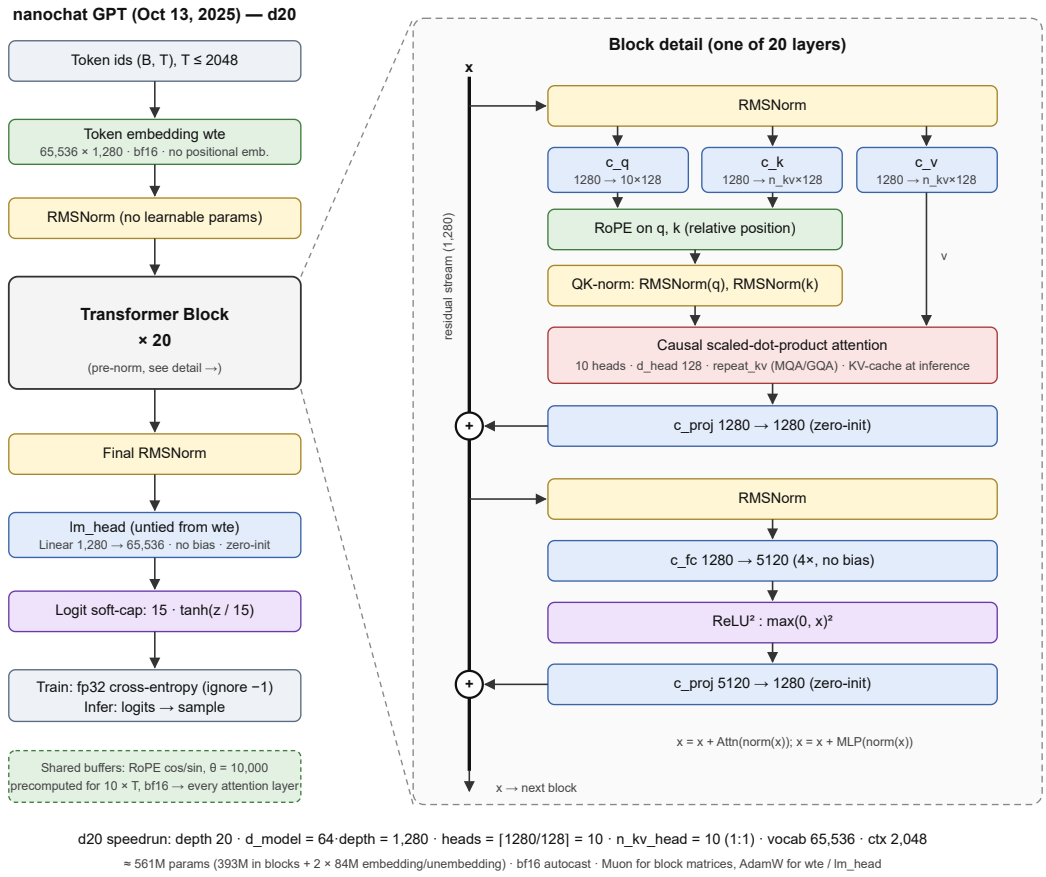


Figure 1: The original nanochat architecture (October 13, 2025) at depth 20. Left: the full model. Right: one of the 20 pre-norm Transformer blocks.

Table 1: Main architecture choices in the original nanochat.

Component	Choice	Why
Position	RoPE on q and k, $\theta = 10{,}000$, all head dims rotated [4]	Relative position with no extra parameters

Component	Choice	Why
Normalization	RMSNorm with no learnable gain, pre-norm, also after the embedding [5]	Cheaper than LayerNorm and keeps the residual stream stable
Attention	Causal multi-head attention with QK-norm [6]	QK-norm bounds attention logits at high learning rates
MLP	$4d$ hidden, ReLU^2 activation [7]	Same cost as GELU and strong in speedruns
Output	Untied `lm_head`; logit soft-cap $15\tanh(z/15)$ [8]	Separate learning rates; prevents over-confident logits
Other	No biases; output projections initialized to zero	Simpler; each block starts as the identity
Optimizer	Muon [9] for the block matrices, AdamW [10] for the embedding and `lm_head`	Muon trains the large matrices faster

Changes in the July 2026 Version

Our baseline is the July 2026 version (`92d63d4`). Most of the changes since October come from the modded-nanogpt speedrun [11]:

- sliding-window attention in an `SSSL` pattern, using FlashAttention-3 [12];
- value embeddings [13] and learned per-layer residual scalars;
- RoPE θ raised to 100,000 and a smaller vocabulary of 32,768 tokens;
- one combined Muon/AdamW optimizer, now with weight decay.

The ReLU^2 MLP has not changed, and RoPE is still applied to every head dimension with one θ for all layers. Our proposed changes target these two parts.

Proposed Changes

ModernBERT [14] updated BERT [15] with components from recent LLMs and tested its choices with ablations. July nanochat already has most of them, including bias-free layers, RoPE, pre-norm, alternating local and global attention, and FlashAttention. We take three ideas from ModernBERT that nanochat lacks: the GeGLU MLP, separate RoPE θ values for local and global layers, and partial RoPE, which ModernBERT tested but did not adopt. Table 2 summarizes them and Figure 2 shows each one.

What we ablate. Part 2 ablates GeGLU and partial RoPE. At d12 neither changes the parameter count or FLOPs, and each is a single flag, so a gap from the baseline that is larger than seed noise can be attributed to the change. We leave change 3 for Part 4: ModernBERT found no difference between θ settings during pretraining and only raised the global θ when it extended its context length. ModernBERT's ablations used encoders of at most 150M parameters, so the results may not carry over to a causal decoder like nanochat.

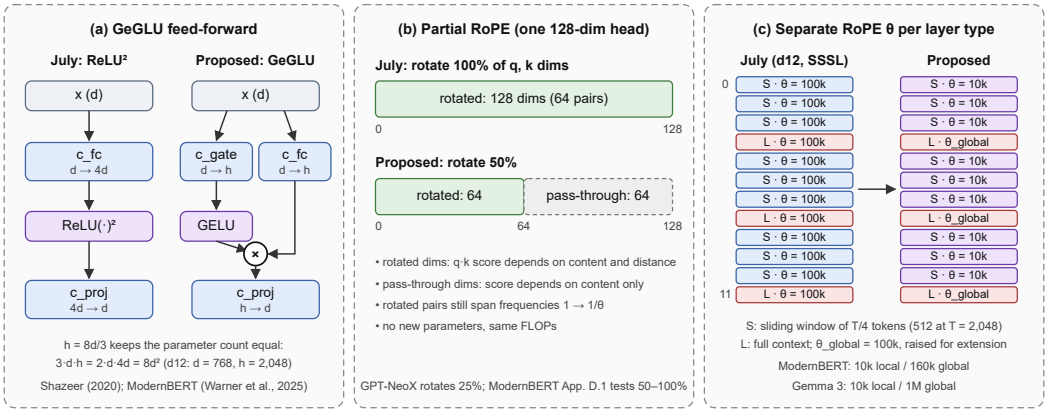


Figure 2: The three proposed changes: (a) GeGLU MLP, (b) partial RoPE, (c) separate RoPE θ for sliding-window (S) and full-context (L) layers.

Table 2: Proposed architecture changes relative to July 2026 nanochat (d = model width, T = context length).

Change	Motivation	Technical Details	Potential Impact
1. GeGLU MLP [16]	In T5, gated MLPs reached lower perplexity than ReLU and GELU MLPs with the same parameter count [16]. PaLM [17], LLaMA [18] and ModernBERT all use one.	$W_{\text{proj}}\big(\text{GELU}(W_{\text{gate}}x) \odot W_{\text{fc}}x\big)$ with $h = 8d/3$, so parameters match ReLU^2 ($h = 2{,}048$ at d12). Flag `--mlp-type=geglu`.	Same parameters and FLOPs. Expect slightly lower loss. ReLU^2 is a strong baseline, so a tie is possible.
2. Partial RoPE [19]	Full RoPE mixes position into every dimension, while unrotated dims can match on content at any distance. GPT-NeoX rotates 25% of each head [19]. ModernBERT found 50–75% slightly better than 100% but kept 100% because its ablations ran at a much smaller scale than the final model.	Rotate the first 64 of the 128 head dims and pass the rest through unchanged. Flag `--rope-frac=0.5`.	No new parameters or FLOPs. Small or no gain at picochat scale; may help context extension.
3. Separate RoPE θ for local and global layers [20]	July uses $\theta = 100{,}000$ everywhere, even in sliding-window layers that only see $T/4$ tokens. ModernBERT uses 10k local and 160k global; Gemma 3 uses 10k and 1M [20].	Two cos/sin tables: sliding-window layers use $\theta = 10{,}000$ and full-context layers keep 100,000. Raise only the global θ when extending context.	No new parameters or FLOPs. We expect no effect at a fixed context length; any benefit should appear during context extension.

Part 2: Ablations on a Nano Nanochat

Part 3: Hardware Efficiency Curves

Part 4: Extending the Context Window

Part 5: Training Our Final Nanochat

References

[1] A. Karpathy, "nanochat: A tiny chatbot arena and training harness," GitHub, 2025. https://github.com/karpathy/nanochat/discussions/481

[2] A. Vaswani et al., "Attention Is All You Need," *NeurIPS*, 2017. arXiv:1706.03762.

[3] A. Radford et al., "Language Models are Unsupervised Multitask Learners," OpenAI technical report, 2019.

[4] J. Su et al., "RoFormer: Enhanced Transformer with Rotary Position Embedding," *Neurocomputing*, vol. 568, 2024. arXiv:2104.09864.

[5] B. Zhang and R. Sennrich, "Root Mean Square Layer Normalization," *NeurIPS*, 2019. arXiv:1910.07467.

[6] A. Henry et al., "Query-Key Normalization for Transformers," *Findings of EMNLP*, 2020. arXiv:2010.04245.

[7] D. R. So et al., "Primer: Searching for Efficient Transformers for Language Modeling," *NeurIPS*, 2021. arXiv:2109.08668.

[8] Gemma Team, "Gemma 2: Improving Open Language Models at a Practical Size," arXiv:2408.00118, 2024.

[9] K. Jordan et al., "Muon: An optimizer for hidden layers in neural networks," 2024. https://kellerjordan.github.io/posts/muon/

[10] I. Loshchilov and F. Hutter, "Decoupled Weight Decay Regularization," *ICLR*, 2019. arXiv:1711.05101.

[11] K. Jordan et al., "modded-nanogpt: Speedrunning the NanoGPT baseline," GitHub, 2024. https://github.com/KellerJordan/modded-nanogpt

[12] J. Shah et al., "FlashAttention-3: Fast and Accurate Attention with Asynchrony and Low-precision," arXiv:2407.08608, 2024.

[13] Z. Zhou et al., "Value Residual Learning For Alleviating Attention Concentration In Transformers," arXiv:2410.17897, 2024.

[14] B. Warner et al., "Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference," *ACL*, pp. 2526–2547, 2025. https://aclanthology.org/2025.acl-long.127/

[15] J. Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," arXiv:1810.04805, 2018.

[16] N. Shazeer, "GLU Variants Improve Transformer," arXiv:2002.05202, 2020.

[17] A. Chowdhery et al., "PaLM: Scaling Language Modeling with Pathways," *JMLR*, vol. 24, 2023. arXiv:2204.02311.

[18] H. Touvron et al., "LLaMA: Open and Efficient Foundation Language Models," arXiv:2302.13971, 2023.

[19] S. Black et al., "GPT-NeoX-20B: An Open-Source Autoregressive Language Model," *BigScience Workshop on Challenges & Perspectives in Creating Large Language Models*, 2022. arXiv:2204.06745.

[20] Gemma Team, "Gemma 3 Technical Report," arXiv:2503.19786, 2025.